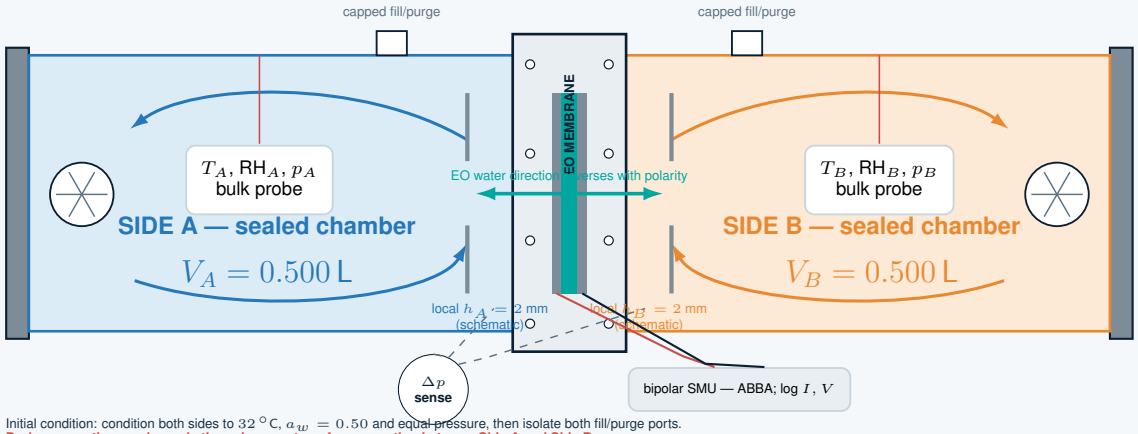
Integrated sealed two-chamber EO vapour cell — configuration and key dimensions

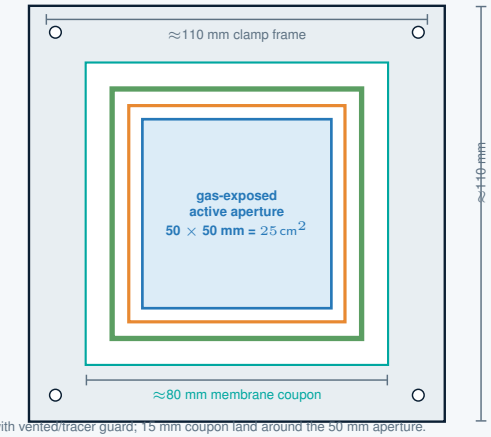
Two calibrated 0.50 L free-gas volumes share one membrane divider; each includes a local 2 mm membrane-face region; no external reservoirs.

A Integrated sealed measurement cell



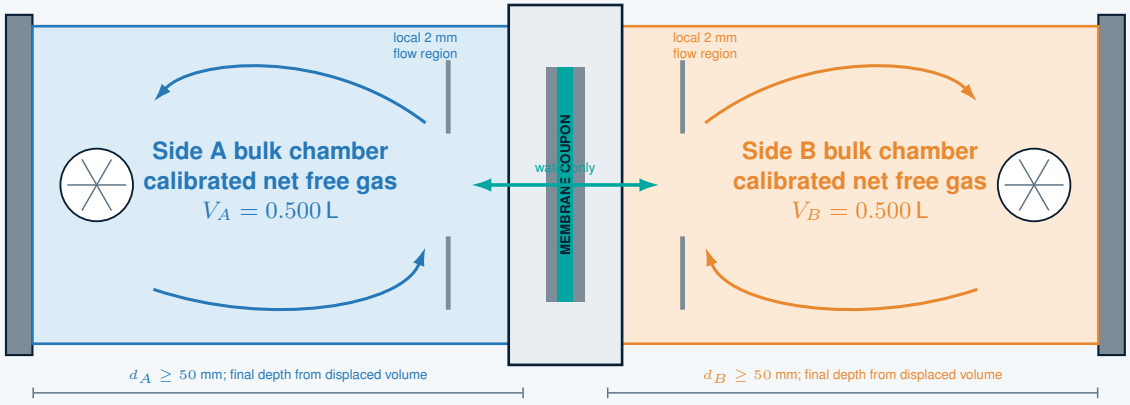
Initial condition: condition both sides to $32\,^{\circ}$C, $a_w = 0.50$ and equal pressure, then isolate both fill/purge ports.
During a run, the membrane is the only mass-transfer connection between Side A and Side B.

B Central divider / membrane cartridge — face view



Double seals with vented/tracer guard; 15 mm coupon land around the 50 mm aperture.
This is the membrane plane; the two chamber depths extend normal to this view.

C Section through the complete integrated two-chamber body

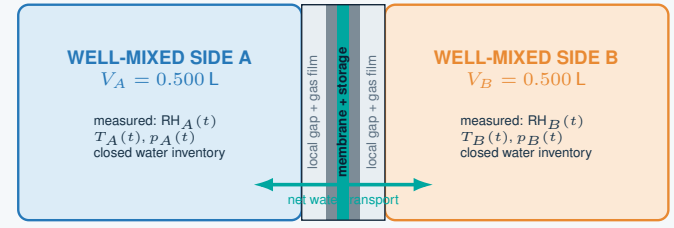


Volume reference per side: 100 × 100 × 50 mm = 0.50 L; increase cavity depth to offset fan, probe, baffle and fitting displacement.

The 2 mm dimension applies only to the 50 × 50 mm membrane-face region (5.00 mL), not to the chamber depth.
Body face concept: ≈140 × 140 mm. Total axial depth is TBD from $d_A + d_B$, end walls and divider; calibrate net volume after assembly.

Concept design, not a manufacturing drawing: the sealed two-chamber transient test is distinct from the once-through steady-flux fixture.

D Model representation and measured states



1. Internal fans homogenise each bulk chamber; they do not add or replace gas.
2. Water lost from one chamber enters the other chamber or is temporarily stored in the membrane.
3. Gas-film resistance links measured bulk RH to the two membrane-surface activities.
4. Differential pressure is monitored but is not used to drive water across the membrane.

No V/Q residence time applies during the sealed run: dynamics are set by gas inventory, membrane transport/storage and interfacial mass transfer.

Report V_A and V_B as measured net free volumes after subtracting all internal hardware displacement.
Validate mixing time, leakage, wall sorption, condensation control and membrane bypass before interpreting RH transients.